Schlenker REV12 — Definitive PLC I/O and Cabinet Construction Basis

Project: Backup Schlenkers 36-10 190036-7-8v2.12 **Engineering platform:** Siemens TIA Portal V19 **PLC:** Siemens S7-1512C-1 PN **Issue date:** 2026-10-10 **Purpose:** definitive PLC address allocation, physical connection destination and cabinet construction baseline.

1. Use of this document

The addresses in Sections 4 and 5 are the approved definitive allocation for the **integrated PLC I/O**. Every signal in Section 4 is wired **directly to the integrated digital-input channel of PLC_1 shown in its first column**. Every signal in Section 5 is wired **directly to the integrated digital-output channel of PLC_1 shown in its first column**. They are not IO-Link signals and shall not be wired to an AL1403 master.

The network areas in Section 5 are allocated in the TIA hardware configuration. Section 6 defines the separate AL1403/IO-Link/SIO architecture. A field device must appear in exactly one architecture. If a device is marked **HOLD — DO NOT WIRE**, the electrical office shall not select a master/port or duplicate a direct-PLC signal.

1.1 Authoritative connection decision

Signals / device	Physical destination	Connection type	Cabinet instruction
`%I10.0`–`%I13.7` listed in Section 4	`PLC_1`, integrated DI, exact channel = address in first column	Direct PLC, 24 VDC digital input	Wire to PLC integrated input terminals; **not AL1403**
`%Q4.0`–`%Q7.7` listed in Section 5	`PLC_1`, integrated DQ, exact channel = address in first column	Direct PLC, 24 VDC digital output	Wire from PLC integrated output terminals; use interface relay where specified; **not AL1403**
`SPL100` IFM `LMT100`	`AL103`, port `P1`; process `%IW324`, PQI `%IB326`	IO-Link	Approved configured port
IFM `IFS240` (`IFK3004BBPKG/US-104`) units	AL1403 port, but exact master/port only after each sensor function and quantity are assigned in Section 6	**SIO digital input**, PNP normally-open; **not IO-Link**	**HOLD — DO NOT WIRE until the assignment table is completed**
Other AL100–AL104 ports	Exact port schedule in Section 7 only	IO-Link or SIO as stated per row	Unassigned ports remain **HOLD — DO NOT WIRE**

Important: the PLC address is not a cabinet terminal number. The electrical office must assign terminal-strip numbers in the schematic, but the destination device and PLC channel above are fixed. Terminal numbers must not be invented in software documentation.

2. Verified automation components

Qty	Tag / device	Component	Interface / role	Construction status
1	PLC_1	Siemens S7-1512C-1 PN compact CPU	PROFINET, 32 integrated DI, 32 integrated DQ	Selected and configured
1	HMI_1	Siemens MTP1200 Unified Comfort as presently shown in the TIA project	PROFINET HMI	Configured; order number to be verified from hardware properties before purchasing
1	G120C_MainMachine	SINAMICS G120C	Main machine VSD, Standard telegram 1	Configured
1	G120C_BottleConveyor	SINAMICS G120C	Bottle conveyor VSD, Standard telegram 1	Configured and bench-tested
1	G120C_ProductPump	SINAMICS G120C	Product pump VSD, Standard telegram 1	Configured
1	G120C_CAP_DISTRIBUTOR	SINAMICS G120C PN, 6SL3210-1KE11-8AF1	Cap distributor VSD, Standard telegram 1	Configured and bench-tested
1	anybus-smc	HMS Anybus ABC3113-A	PROFINET to EtherCAT gateway	Selected and configured
1	Valve manifold	SMC EX260-SEC1 with eight VQC1100N-51 valves	EtherCAT pneumatic output manifold	Selected; final valve/coil wiring to follow pneumatic drawings
5	AL100–AL104	IFM AL1403 IO-Link masters	PROFINET field I/O	Selected and configured
1	SPL100	IFM LMT100	Product/water presence before pump, AL103 port 1	Detected and mapped
1	M100	Main machine motor	Driven by G120C	Motor data to be verified before final VSD commissioning
1	M101	Bottle conveyor motor	Driven by G120C	Motor data to be verified before final VSD commissioning
1	M102	Product pump motor, ABB M2QA90L2A	2.2 kW, 380–420 V Y, 4.69 A, 50 Hz, 2850 rpm	Plate data recorded
1	M103	Cap distributor motor, Western Electric	0.37 kW, 415 V, 50 Hz, 1370 rpm	Bench-tested; rated current and cos phi still require clear plate confirmation

3. Project tree — automation, drives and field I/O

This tree is the construction view of the project. It separates direct PLC I/O, PROFINET devices, the Anybus/SMC pneumatic branch and the AL1403 field-sensor branch.

Parent	Child / device	Interface	Connected functions
Project	PLC_1 S7-1512C-1 PN	Main controller	Program, integrated DI/DQ and PROFINET controller
PLC_1	Integrated DI %I10.0–%I13.7	Direct 24 VDC inputs	All Section 4 input signals; not IO-Link
PLC_1	Integrated DQ %Q4.0–%Q7.7	Direct 24 VDC outputs	Lights, filter relay, requests and Section 5 outputs; not IO-Link
PLC_1	HMI_1 MTP1200 Unified Comfort	PROFINET	Operator commands, status, alarms and setpoints
PLC_1	G120C_MainMachine	PROFINET / telegram 1	Main-machine motor
PLC_1	G120C_BottleConveyor	PROFINET / telegram 1	Bottle-conveyor motor
PLC_1	G120C_ProductPump	PROFINET / telegram 1	Product-pump motor
PLC_1	G120C_CAP_DISTRIBUTOR	PROFINET / telegram 1	Cap-distributor motor
PLC_1	anybus-smc HMS ABC3113-A	PROFINET gateway	Four-byte command image %QD128
anybus-smc	SMC EX260-SEC1 valve manifold	EtherCAT	Eight VQC1100N-51 pneumatic valves; final coil/function schedule follows pneumatic design
PLC_1	AL100	PROFINET IO-Link master	Ports remain HOLD until verified field allocation
PLC_1	AL101	PROFINET IO-Link master	Ports remain HOLD until verified field allocation
PLC_1	AL102	PROFINET IO-Link master	Ports remain HOLD until verified field allocation
PLC_1	AL103	PROFINET IO-Link master	P1 = SPL100/LMT100; remaining ports HOLD
PLC_1	AL104	PROFINET IO-Link master	Ports remain HOLD until verified field allocation
AL100–AL104	IFM IFS240 sensors	M12, SIO DI, PNP NO	Exact sensor quantity/function/master/port still required; DO NOT WIRE until allocated

3.1 Available verified photographs



SINAMICS G120C photographed on the bench



IFM IFS240 package photographed by the operator



Cap distributor motor nameplate photographed by the operator

Only photographs that identify the actual supplied or bench-tested component are used. Components without an operator photograph remain identified by verified tag and model in the tree; no catalogue image of a possibly different variant is substituted.

4. Integrated digital inputs — definitive allocation

PHYSICAL DESTINATION FOR EVERY ROW BELOW: `PLC_1` **integrated DI, direct 24 VDC input. The exact input channel is the** `%I...` **address in the first column. Do not connect these signals to AL100–AL104.**

PLC address	PLC signal	Field function	Electrical drawing requirement
%I10.0	DB_Global.Inp.SafetyOK	Safety chain healthy	Potential-free/24 VDC status from safety system; not a replacement for safety-rated wiring
%I10.1	DB_Global.Inp.AirPressureOK	Main pneumatic pressure healthy	Digital sensor/contact
%I10.2	DB_Global.Inp.EncoderHealthy	Main encoder healthy	Diagnostic contact/interface status
%I10.3	DB_Global.Inp.MachineZero	Machine zero/reference position	Digital sensor
%I10.4	DB_Global.Inp.BottleInfeed	Bottle detected at infeed	Photoelectric sensor
%I10.5	DB_Global.Inp.TankFull100	Tank full / 100% limit	Level switch/status
%I10.6	DB_Global.Inp.BottleShortage1	Bottle shortage stage 1	Digital sensor/status
%I10.7	DB_Global.Inp.BottleShortage2	Bottle shortage stage 2	Digital sensor/status
%I11.0	DB_Global.Inp.Accumulation1	Bottle accumulation stage 1	Digital sensor/status
%I11.1	DB_Global.Inp.Accumulation2	Bottle accumulation stage 2	Digital sensor/status
%I11.2	DB_Global.Inp.GateOpenFB	Gate open feedback	Limit/reed switch
%I11.3	DB_Global.Inp.GateClosedFB	Gate closed feedback	Limit/reed switch
%I11.4	DB_Global.Inp.FillerLowLimit	Filler lower limit	Limit switch
%I11.5	DB_Global.Inp.FillerHighLimit	Filler upper limit	Limit switch
%I11.6	DB_Global.Inp.CapperLowLimit	Capper lower limit	Limit switch
%I11.7	DB_Global.Inp.CapperHighLimit	Capper upper limit	Limit switch
%I12.0	DB_Global.Inp.CapPresent	Cap present	Cap sensor
%I12.1	DB_Global.Inp.CapHopperLow	Cap hopper low	Level sensor
%I12.2	DB_Global.Inp.CapChannelDemand	Cap channel demand	Digital sensor/status
%I12.3	DB_Global.Inp.CapsMissing	Caps missing / chute empty	Digital sensor/status
%I12.4	DB_Global.Inp.HoodFilterAlarm	Hood/filter alarm	Differential-pressure/status contact
%I12.5	DB_Global.Inp.ControlPowerOK	24 VDC control power healthy	PSU/monitor auxiliary contact
%I12.6	DB_Global.Inp.AirFilterPressureOK	Air-filter pressure proof	Pressure-switch contact
%I12.7	DB_Global.Inp.AirFilterChangeRequired24V	Filter replacement required	Filter monitor contact
%I13.0	DB_Global.Inp.RequestOpenDoorPanelPB	Panel-door request pushbutton	24 VDC pushbutton contact
%I13.1	DB_Global.Inp.RequestOpenDoorFrontPB	Front-door request pushbutton	24 VDC pushbutton contact
%I13.2	DB_Global.Inp.RequestOpenDoorRearPB	Rear-door request pushbutton	24 VDC pushbutton contact
%I13.3	DB_Global.Inp.ControlledStopComplete	Controlled stop complete	External sequence/status contact
%I13.4	DB_Global.Inp.ZeroSpeedConfirmed	Zero speed confirmed	Drive/safety status contact
%I13.5	DB_Global.Inp.ThreePhaseOffConfirmed	Three-phase supply isolated confirmation	Contactor/monitor auxiliary contact
%I13.6	DB_Global.Inp.AllDoorsClosed	All guarded doors closed	Safety-system status only; safety function remains hardwired/safety-rated
%I13.7	DB_Global.Inp.ResetAlarmPanelPB	Panel alarm reset pushbutton	24 VDC pushbutton contact

5. Integrated digital outputs — definitive allocation

PHYSICAL DESTINATION FOR EVERY ROW BELOW: `PLC_1` **integrated DQ, direct 24 VDC output. The exact output channel is the** `%Q...` **address in the first column. Do not connect these signals to AL100–AL104.**

PLC address	PLC signal	Field function	Electrical drawing requirement
%Q4.0	DB_Global.Out.VacuumPump	Vacuum-pump contactor command	Interposing relay/contactor circuit; protection and overload feedback per motor design
%Q4.1	DB_Global.Out.MachineLightsRelayCmd	Machine-lights relay command	Interposing relay for lighting circuit
%Q4.2	DB_Global.Out.AirFilterRelayCmd	Air-filter relay command	Interposing relay/contactor for filter unit
%Q4.3	DB_Global.Out.GateOpen	Gate-open solenoid command	Valve/relay circuit
%Q4.4	DB_Global.Out.GateClose	Gate-close solenoid command	Valve/relay circuit
%Q4.5	DB_Global.Out.FillerUp	Filler lift up command	Valve/relay circuit
%Q4.6	DB_Global.Out.FillerDown	Filler lift down command	Valve/relay circuit
%Q4.7	DB_Global.Out.CapperUp	Capper lift up command	Valve/relay circuit
%Q5.0	DB_Global.Out.CapperDown	Capper lift down command	Valve/relay circuit
%Q5.1	DB_Global.Out.EV210	Solenoid valve EV210	Valve coil/relay circuit
%Q5.2	DB_Global.Out.EV211	Solenoid valve EV211	Valve coil/relay circuit
%Q5.3	DB_Global.Out.EV212	Solenoid valve EV212	Valve coil/relay circuit
%Q5.4	DB_Global.Out.EV220	Solenoid valve EV220	Valve coil/relay circuit
%Q5.5	DB_Global.Out.EV221	Solenoid valve EV221	Valve coil/relay circuit
%Q5.6	DB_Global.Out.EV230FastRelease	EV230 fast-release command	Valve coil/relay circuit
%Q5.7	DB_Global.Out.ExternalBottleWashValveCmd	External bottle-wash valve	Valve coil/relay circuit
%Q6.0	DB_Global.Out.UnlockAllDoorsRequest	Guard-door unlock request	Request to safety/locking circuit; never use as the safety function itself
%Q6.1	DB_Global.Out.RequestPanelBlueLampCmd	Panel request blue lamp	24 VDC lamp/relay circuit
%Q6.2	DB_Global.Out.RequestFrontBlueLampCmd	Front request blue lamp	24 VDC lamp/relay circuit
%Q6.3	DB_Global.Out.RequestRearBlueLampCmd	Rear request blue lamp	24 VDC lamp/relay circuit
%Q6.4	DB_Global.Out.Valve217Cmd	Valve 217 command	Valve coil/relay circuit
%Q6.5	DB_Global.Out.Valve213Cmd	Valve 213 command	Valve coil/relay circuit
%Q6.6	DB_Global.Out.Valve212Cmd	Valve 212 command	Valve coil/relay circuit
%Q6.7	DB_Global.Out.Valve247Cmd	Valve 247 command	Valve coil/relay circuit
%Q7.0	DB_Global.Out.InternalWashValveCmd	Internal-wash valve command	Valve coil/relay circuit
%Q7.1	DB_Global.Out.CIPReqColdWater	CIP cold-water request	Interface relay/status output
%Q7.2	DB_Global.Out.CIPReqHotWater	CIP hot-water request	Interface relay/status output
%Q7.3	DB_Global.Out.CIPReqAcid	CIP acid request	Interface relay/status output
%Q7.4	DB_Global.Out.CIPReqCitra	CIP Citra request	Interface relay/status output
%Q7.5	DB_Global.Out.CIPReqDischarge	CIP discharge request	Interface relay/status output
%Q7.6	DB_Global.Out.ProductionReqProduct	Production product request	Interface relay/status output
%Q7.7	Reserved, forced FALSE	Spare integrated output	Bring to spare terminal; keep unenergised in software

6. PROFINET and distributed I/O areas

Device / area	PLC input area	PLC output area	Function
G120C main machine	%IW308, %IW310	%QW140, %QW142	Status word, actual speed; control word, speed setpoint
G120C bottle conveyor	%IW312, %IW314	%QW144, %QW146	Status word, actual speed; control word, speed setpoint
G120C product pump	%IW316, %IW318	%QW148, %QW150	Status word, actual speed; control word, speed setpoint
G120C cap distributor	%IW320, %IW322	%QW152, %QW154	Status word, actual speed; control word, speed setpoint
HMS Anybus ABC3113-A	Device diagnostics	%QD128	Four-byte command image to SMC EX260-SEC1
IFM AL104	36-byte input area beginning %IB126	%QB132	IO-Link master process/diagnostic area; port functions per final device schedule
IFM AL102	36-byte input area beginning %IB162	%QB133	IO-Link master process/diagnostic area; port functions per final device schedule
IFM AL101	36-byte input area beginning %IB198	%QB134	IO-Link master process/diagnostic area; port functions per final device schedule
IFM AL100	36-byte input area beginning %IB234	%QB135	IO-Link master process/diagnostic area; port functions per final device schedule
IFM AL103	36-byte input area beginning %IB270	%QB136	IO-Link master; SPL100/LMT100 process word at %IW324, PQI at %IB326

7. IFM AL1403 port allocation and IFS240 connection

7.1 Approved configured device

Device	Function	Master	Port	Mode	PLC process data	Status
SPL100 / IFM LMT100	Product/water presence before pump	AL103	P1	IO-Link	%IW324; PQI %IB326	**APPROVED / CONFIGURED**

7.2 IFM IFS240 sensors shown by the operator

The photographed package identifies IFM IFS240, order code IFK3004BBPKG/US-104. Manufacturer data identifies it as a three-wire, 10–30 VDC, PNP normally-open inductive sensor with M12 connector. It has **no IO-Link interface**. When used through an AL1403 it must therefore use **SIO digital-input mode**, not IO-Link/IODD mode.

Manufacturer reference: https://media.ifm.com/dam/f1582b25-9600-4283-9368-1fd3025d6291/Original/IFS240-01_EN-GB.pdf

Sensor instance	Machine function	Master	Port	Mode	PLC tag/address	Construction status
IFS240-01...n	Quantity and individual function have not yet been supplied	**NOT ASSIGNED**	**NOT ASSIGNED**	AL1403 **SIO DI**, PNP NO	**NOT ASSIGNED**	**HOLD — DO NOT WIRE**

The office must receive the quantity and one machine-function name for each physical IFS240 before master/port allocation. Assigning AL100–AL104 ports without that information would invent the wiring and may duplicate a signal already fixed to the direct PLC inputs in Section 4.

7.3 AL1403 port rule

- A port is usable only when its row states device, machine function, master, port, mode and PLC tag/address.
- IO-Link is used only for a verified IO-Link-capable device with the correct port process-data configuration.
- SIO DI is used for a compatible conventional switching sensor such as IFS240.
- No signal from Section 4 may be repeated on an AL1403 port unless this document and the PLC program are formally revised together.
- The earlier planning allocation that associated bottle/cap functions with AL100/AL101 is **not a wiring authorization** because those functions currently exist in the definitive direct-PLC schedule.

8. Cabinet construction rules

1. Provide fused/protected 24 VDC distribution and separate field-power groups according to the final load calculation and applicable electrical standard.
2. Use interposing relays/contactors wherever the field load exceeds the PLC output rating or requires galvanic separation. Confirm every coil current before selecting the relay.
3. Fit suppression appropriate to every inductive DC coil. Confirm polarity where suppression devices are polarised.
4. Keep safety functions in the approved safety architecture. Standard PLC inputs and outputs in this schedule provide status and requests only; they do not replace safety-rated devices or circuits.
5. Terminate `%Q7.7` as a labelled spare and leave it unenergised.
6. Label wires, terminals and field devices with the PLC address and the tag in this document.
7. Keep PROFINET, motor, encoder, analogue and low-level sensor cables segregated from power wiring according to the manufacturer instructions.
8. Before energisation, perform point-to-point continuity, insulation, polarity, protective-earth and 24 VDC short-circuit checks.

9. Items the electrical office must complete before purchasing

- Where an exact order number is not stated in Section 2, obtain it from the approved TIA hardware properties or final vendor quotation; do not derive it from the description alone.
- Confirm the exact HMI order number because current project evidence shows MTP1200 Unified Comfort while older REV12 documentation names MTP1500 Unified Comfort.
- Export the exact CPU and accessory order numbers from TIA hardware properties.
- Size the 24 VDC power supply, protective devices, interposing relays, contactors, terminals and cable cross-sections from the final load calculation.
- Confirm motor protection and cable data for M100, M101 and M103 from clear motor/VSD nameplates.
- Provide the quantity and machine function of every IFS240, then formally allocate each unit to an AL1403 master/port in SIO mode before wiring.
- Finalise the IO-Link/SIO port allocation for all devices other than SPL100/LMT100 on AL103/P1.
- Produce the safety schematic and validate the safety performance level independently of the standard PLC allocation above.

10. Software verification

The definitive integrated I/O assignments are implemented in `FB_Application`. All 32 integrated outputs are cleared on CPU startup in `OB100_Startup`. The TIA V19 import and PLC compilation completed with **0 errors and 0 warnings** on 2026-10-09.